Benchmarking Multi-View Tree-Level Oil Palm Bunch Counting Under a Fixed Detector

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Abstract—Black Bunch Census (BBC) supports oil palm harvest planning by estimating, for every tree, how many fresh fruit bunches (FFB) sit in each operational maturity class. Image-based BBC is difficult because a single tree is observed from multiple side views: additional views reduce occlusion, but the same physical bunch can appear in several images and inflate naive counts. Despite an active literature on agricultural detection and counting, no benchmark to date has separated detector error from tree-level counter error for this task. This paper benchmarks multi-view tree-level FFB counting for B1, B2, B3, and B4 under two carefully isolated detection conditions. In the GT detection setting, counters receive ground-truth boxes and classes, removing detector quality as a source of error. In the fixed-detector setting, counters receive cached YOLOv26-medium outputs, measuring the deployed detector-plus-counter pipeline. The benchmark uses 953 trees, 3,992 side-view images, four maturity classes, 4 to 8 views per tree, and a fixed 716/96/141 train/validation/test split. Under GT detection, ElasticNet reaches 98.05% Class ±1 Acc and 92.20% Tree ±1 Acc. Under the fixed detector, the best Ridge + F_{all} pipeline reaches 77.48% Class ±1 Acc and 32.62% Tree ±1 Acc. Feature ablation closes only ≈1.4 pp Class ±1 Acc over the F0 baseline, and changing the counter family does not recover the gap either. The conclusion is direct: tree-level counting is near ceiling when detections are accurate, and the remaining 20.57 pp gap is a detector-quality bottleneck, not a counter limitation.

Index Terms—Oil palm, fresh fruit bunch, black bunch census, multi-view counting, object detection, YOLO, tree-level regression, agricultural computer vision.

I. Introduction

Oil palm harvest planning is driven by Black Bunch Census (BBC), a field practice in which counters estimate, for every tree, how many bunches belong to each operational maturity class: typically B1 (ripe, harvest now), B2 (imminent), B3 (next cycle), and B4 (future inventory) [11]. The decision-relevant signal is therefore not the total number of bunches on a tree, but the four-dimensional count vector across maturity classes. Image-based BBC must reproduce this per-class disaggregation rather than collapse it into a single total.

Tree-level counting is geometrically harder than image-level detection [12]. Bunches surround the full circumference of an oil palm, so a single image misses bunches because of occlusion, viewing angle, or overlapping fronds. Multi-view acquisition reduces missed bunches by photographing a tree from four or eight side views, but the same physical bunch then appears in several of those images. Fig. 1 illustrates the effect: one B3 bunch is visible, and would be detected, in three adjacent side views of tree DAMIMAS_A21B_0847. A naive sum over per-image detections therefore counts appearances, not bunch identities, and is biased upward by roughly a factor of two on this dataset.

Three lines of prior work bear on this problem. Oil palm FFB detection and ripeness classification have been studied with both two-stage and YOLO-style detectors [8]–[10], [22]; these works show that detectors can reach usable mAP on FFB but also document recurring weaknesses on partially occluded or maturity-ambiguous bunches.

Multi-view fruit aggregation has been used in fruit recognition and yield estimation, including methods that explicitly fuse evidence across views to avoid double counting [5], [6]. Comparative studies on apple orchards and apple-and-orange counting from robotic platforms have further refined these pipelines [14], [15]. These methods, however, are evaluated mainly on smaller tree-fruit crops with simpler geometry than oil palm, and they do not isolate detector quality from aggregator quality.

Regression-based counting and yield estimation in agricultural vision are well established [1], [2], [7], [12], [13], and YOLO-style single-stage detectors are now standard front-ends for such pipelines [3], [4], [17], [18], with deep two-stage detectors also widely applied in earlier orchard work [16]. In this line of work, a regressor maps aggregate detection statistics to a target count, but the regressor is typically evaluated only on the detector outputs at hand, leaving the contribution of detector quality versus regressor quality unmeasured.

What is missing is a benchmark that determines, for tree-level multi-view BBC counting, whether the limiting factor lies in the detector or in the tree-level aggregator. Existing reports usually fix one component and tune the other, conflating two error sources. This paper closes that gap. The contributions are:

1) A systematic benchmark of tree-level bunch counting under two detection conditions. Counter models are evaluated under GT detections (upper bound on counter performance) and under cached YOLOv26-medium outputs (deployed pipeline), with the same dataset split and the same evaluation metrics.

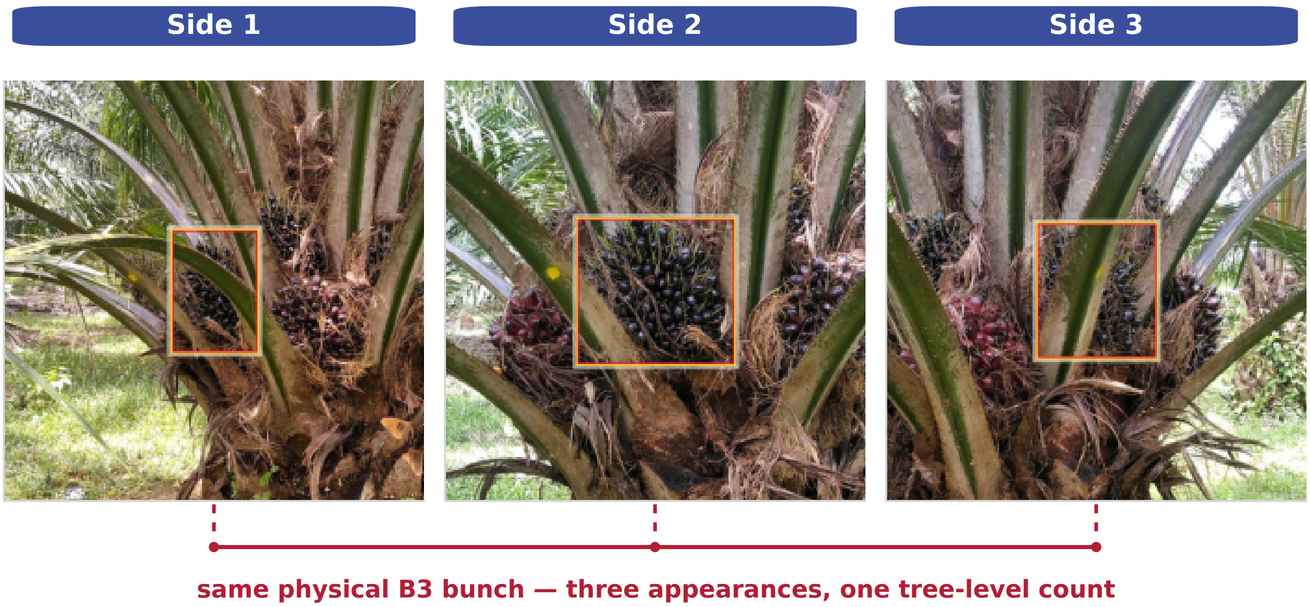

Fig. 1. A single physical B3 bunch on tree DAMIMAS_A21B_0847 is visible across sides 1, 2, and 3. If every per-image detection is counted, three appearances are added for one bunch. Tree-level counting must therefore aggregate evidence across views rather than sum it.

2) A feature ablation under realistic detector outputs. Eight feature banks ranging from a 13-dimensional baseline to a 67-dimensional combined bank are evaluated with Ridge regression to identify which feature groups carry counting signal once detector noise is present.
3) Evidence that detector output quality is the primary bottleneck, not the counter. The two settings differ by 20.57 percentage points in Class ±1 Acc, and feature richness or counter choice closes only a small fraction of that gap.

II. Methodology

A. Dataset and Task Definition

The benchmark uses the SawitMVC-YOLO multi-view dataset: 953 oil palm trees, 3,992 side-view images, four maturity classes (B1, B2, B3, and B4), and 4 to 8 side views per tree. The fixed split contains 716 training trees, 96 validation trees, and 141 test trees. Dataset construction, annotation protocol, and per-class composition are documented in the companion dataset release.

For tree i, the ground-truth target is the count vector

$$\mathbf{y}_i = [\, y_{i,B1},\; y_{i,B2},\; y_{i,B3},\; y_{i,B4}\,], \tag{1}$$

and the prediction is $\hat{\mathbf{y}}_i = [\,\hat{y}_{i,B1}, \ldots, \hat{y}_{i,B4}\,]$. The evaluation unit is the tree, not the individual image: per-image appearance counts are not valid BBC outputs.

Let $a_{i,c}$ denote the number of detected appearances of class c across all views of tree i. The naive multi-view estimator

$$\hat{y}_{i,c}^{\text{naive}} = a_{i,c} \tag{2}$$

is biased upward whenever a bunch is visible from more than one side. Under GT annotations, this estimator reaches only 50.00% Class ±1 Acc and 6.38% Tree ±1 Acc on the test split, which motivates learned tree-level aggregation rather than direct summation.

B. Detection Conditions

The benchmark isolates detector error from counter error by evaluating the same counters under two detection conditions, illustrated in Fig. 2.

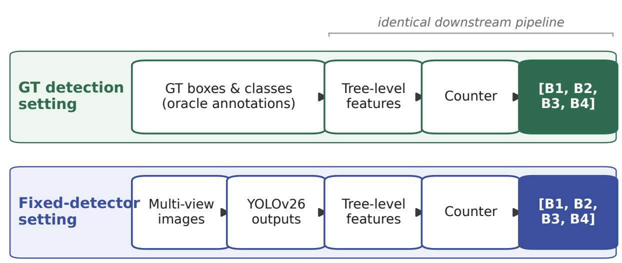

Fig. 2. Two detection conditions used in the benchmark. Top: the GT detection setting supplies ground-truth boxes and classes to the tree-level feature extractor, so detector output quality is removed as an error source. Bottom: the fixed-detector setting supplies cached YOLOv26-medium outputs, so detector misses and class confusions enter the pipeline before counting.

In the GT detection setting, the feature extractor reads ground-truth bounding boxes and classes directly from the annotation files. Counters trained on these features upper-bound how well a tree-level aggregator can perform when detections are accurate. In the fixed-detector setting, the feature extractor reads cached YOLOv26-medium [18] predictions. A single detector checkpoint is used for all counter and feature configurations so that any observed differences are attributable to the counter or features, not to detector retraining.

TABLE I
YOLOv26-medium validation detection performance, reported under the COCO mAP50 convention [19].

Class	mAP50	Recall
B1	0.746	0.801
B2	0.425	0.433
B3	0.550	0.656
B4	0.363	0.389
Overall	0.521	0.570

The detector is strongest on B1 and weakest on B4. The per-class weakness on B4 and the moderate recall on B3 visible in Table I are revisited in Section III-D, where the per-class structure of the fixed-detector gap mirrors this profile.

C. Feature Vectors

For each tree, detections from all views are aggregated into a fixed-length feature vector. The 13-dimensional baseline F0 contains, for each class c, the total count s_c, the per-side maximum m_c, and the per-side mean μ_c, plus the number of available views n_{views}:

$$F0 = [\, s_{B1:B4},\ m_{B1:B4},\ \mu_{B1:B4},\ n_{\text{views}}\,]. \tag{3}$$

Three optional groups extend F0:

- Confidence (20-dim): five statistics per class (confidence sum, mean, max, count above 0.5, and count above 0.6), summarizing how strongly detections are scored.
- Spatial (8-dim): two statistics per class (mean normalized vertical centroid $\overline{cy}_c$ and mean bounding-box area $\overline{A}_c$), capturing the fact that maturity classes tend to occupy distinct vertical zones on the tree.
- Side-distribution (20-dim): five statistics per class (per-side std, per-side min, coefficient of variation, number of sides with detections, and a consistency score), capturing how detections spread across the available views.

A cross-class composition group (6-dim), consisting of the total detection count, four class fractions, and a B3-vs-(B2+B3) mixture ratio, completes the 67-dimensional combined bank F_{all}:

$$F_{\text{all}} = F0 \cup \text{conf} \cup \text{spatial} \cup \text{distrib} \cup \text{composition}.$$

Eight feature banks are evaluated: F0, F0+conf, F0+spatial, F0+distrib, F0+conf+spatial, F0+conf+distrib, F0+distrib+spatial, and F_{all}.

D. Counter Models and Evaluation Metrics

Five regression models are evaluated in both detection conditions: Linear Regression, Ridge [23], ElasticNet [20], SVM [24], and Random Forest [21]. Each counter maps a tree feature vector $\mathbf{x}_i$ to the count vector $\hat{\mathbf{y}}_i = f_\theta(\mathbf{x}_i)$.

All counter models are trained on the 716-tree training split and evaluated on the fixed 141-tree test split. Class-level ±1 correctness for tree i and class c is

$$I^{\pm1}_{i,c} = \begin{cases} 1, & |\hat{y}_{i,c} - y_{i,c}| \le 1, \\ 0, & \text{otherwise.} \end{cases} \tag{4}$$

Class ±1 Acc averages this indicator over all test trees and classes:

$$\text{Class } \pm 1 \text{ Acc} = \frac{1}{4N} \sum_{i=1}^{N} \sum_{c=1}^{4} I^{\pm1}_{i,c}. \tag{5}$$

Tree ±1 Acc is stricter: a tree counts as correct only if all four class predictions are within ±1 simultaneously,

$$\text{Tree } \pm 1 \text{ Acc} = \frac{1}{N} \sum_{i=1}^{N} \mathbb{1}\left(\sum_{c=1}^{4} I^{\pm1}_{i,c} = 4 \right). \tag{6}$$

Macro MAE is the mean absolute error averaged over trees and classes. Signed class bias is the per-class mean $\hat{y}_{i,c} - y_{i,c}$; positive values indicate over-counting, negative values under-counting. Bias is reported alongside accuracy because operational aggregate yield estimates are sensitive to directional error, not just absolute error.

III. Results and Discussion

The results are arranged to follow the argument: the ceiling first (Section III-A), the reality under a real detector (Section III-B), the feature ablation that rules out the counter side (Section III-C), and the gap analysis that locates the bottleneck (Section III-D).

A. GT Detection Setting

Table II reports counting results when the counter receives GT detections. Naive appearance summation fails on the test set. This confirms that duplicate visibility cannot be ignored. Both heuristic divisor rules quickly recover to above 95% Class ±1 Acc, and learned counters push this to 98.05% with ElasticNet.

TABLE II
Counting under the GT detection setting on the 141-tree test split.

Method	Type	Class ±1	Tree ±1	MAE
Naive sum	Heur.	50.00%	6.38%	2.142
Global divisor	Heur.	95.39%	85.11%	0.376
Visibility-adapt. divisor	Heur.	95.92%	87.23%	0.340
ElasticNet	ML	98.05%	92.20%	0.277
SVM	ML	97.87%	91.49%	0.266
Ridge	ML	97.70%	90.78%	0.275
Linear reg.	ML	97.52%	90.07%	0.277
Random forest	ML	95.92%	84.40%	0.365

The four best machine-learning counters (ElasticNet, SVM, Ridge, LR) sit within 0.5 percentage points of each other on Class ±1 Acc and within 2.2 points on Tree ±1 Acc; Random Forest is the only counter that underperforms in this setting. The counter formulation, then, is not the limiting factor: when detector evidence is accurate, simple regularized linear models suffice to recover the BBC count.

B. Fixed-Detector Setting

Table III holds the feature bank at F0 and compares the five counter families when they receive cached YOLOv26-medium outputs. The best F0 result, ElasticNet at 76.42% Class ±1 Acc, is more than 20 percentage points below the ElasticNet result of the GT detection setting under the same feature bank. All five counters fall within a 3.2 percentage-point window on Class ±1 Acc, reinforcing that the counter family is not what changed between the two settings; only the detector did.

TABLE III
Fixed-detector counting with F0 features on the 141-tree test split.

Counter	Class ±1 Acc	Tree ±1 Acc	Macro MAE
ElasticNet	76.42%	29.79%	1.043
Ridge	76.06%	28.37%	1.053
Linear Regression	75.71%	30.50%	1.048
SVM	74.82%	29.08%	1.043
Random Forest	73.23%	26.95%	1.110

Per-class accuracies, listed in Table IV, reveal where the loss concentrates: B1 stays above 95% across all five counters, B2 sits between 73.76% and 80.14%, and B3 collapses to between 55% and 56%. B4 is intermediate, between 67% and 73%. This per-class profile mirrors the per-class detector performance in Table I, where B3 has moderate recall and B4 has the lowest recall.

TABLE IV
Per-class Class ±1 Acc of fixed-detector counters with F0 features on the 141-tree test split.

Counter	B1	B2	B3	B4
ElasticNet	96.45%	80.14%	56.03%	73.05%
Ridge	95.74%	80.14%	56.03%	72.34%
Linear Regression	96.45%	79.43%	55.32%	71.63%
SVM	95.74%	76.60%	55.32%	71.63%
Random Forest	95.74%	73.76%	56.03%	67.38%

C. Feature Ablation

TABLE V
Ridge feature ablation in the fixed-detector setting.

Features	Dim	Class ±1	Tree ±1	MAE
F0	13	76.06%	28.37%	1.053
F0 + confidence	33	75.89%	29.79%	1.059
F0 + spatial	21	76.60%	30.50%	1.046
F0 + side-distribution	33	74.82%	28.37%	1.060
F0 + confidence + spatial	41	76.24%	29.08%	1.059
F0 + confidence + side-distribution	53	76.42%	29.79%	1.060
F0 + side-distribution + spatial	41	75.71%	30.50%	1.048
F_{all}	67	77.48%	32.62%	1.035

Table V holds the counter family fixed at Ridge (the strongest counter under the richest feature bank) and varies only the feature configuration. Spatial features alone bring 0.54 percentage points over F0, confidence and side-distribution alone do not help, and the full F_{all} bank reaches 77.48% Class ±1 Acc and 32.62% Tree ±1 Acc. The headline absolute gain from F0 to F_{all} is +1.42 percentage points in Class ±1 Acc and +4.25 percentage points in Tree ±1 Acc. This is the largest improvement that any feature ablation produces in the fixed-detector setting, and it is small relative to the 20.57 percentage points lost between the two settings. Feature engineering, therefore, can only marginally compensate for what the detector did not supply.

D. GT vs. Fixed-Detector Gap

Fig. 3 visualises the central comparison. The top panel shows per-class Class ±1 Acc under both detection conditions; the bottom panel shows the per-class signed bias of the best fixed-detector configuration.

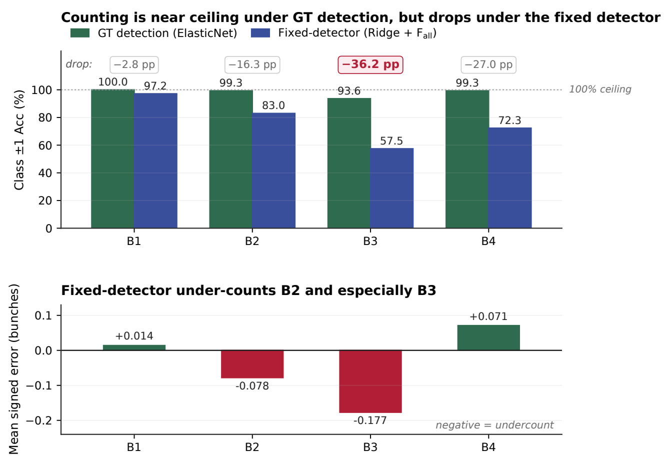

Fig. 3. Top: Per-class Class ±1 Acc for the best counter in each detection condition (ElasticNet under GT; Ridge + F_{all} under the fixed detector). Counting is near ceiling under GT detection but drops sharply under the fixed detector. Bottom: Per-class signed bias of Ridge + F_{all} in the fixed-detector setting. Negative values indicate systematic under-counting; the under-count is concentrated on B2 and especially B3.

Three features of Fig. 3 stand out. The absolute gap is large, at 20.57 percentage points in Class ±1 Acc and 59.58 percentage points in Tree ±1 Acc, summarised in Table VI. Beyond its magnitude, the gap has per-class structure. B1 loses only ≈2.8 pp between the two conditions, B2 loses ≈16.3 pp, B3 loses 36.2 pp, and B4 loses ≈27.0 pp. This profile aligns with the per-class detector performance in Table I, where the classes with weaker recall (B3, B4) fall furthest below the GT ceiling. The bias is directional. The fixed-detector Ridge + F_{all} pipeline under-counts B2 by 0.078 and B3 by 0.177 bunches per tree on average. Because plantation-level yield estimates aggregate per-tree predictions across blocks, a systematic per-class undercount in B2 and B3 propagates into a systematic underestimate of the next-cycle harvest pipeline.

TABLE VI
Consolidated test-set summary. Biases are ordered by class.

Config	Class ±1	Tree ±1	B1	B2	B3	B4
GT, ElasticNet	98.05%	92.20%	−.050	+.043	−.064	−.028
Fixed, Ridge + F_{all}	77.48%	32.62%	+.014	−.078	−.177	+.071

Read together, Tables I–VI with Fig. 3 support a single interpretation. Multi-view duplicate visibility is real and large, but it is well handled by learned tree-level aggregation when detector evidence is accurate (Section III-A). Once a fixed real-world detector replaces ground-truth boxes, counter performance collapses (Section III-B), and richer features or different counter families recover only a small fraction of the loss (Section III-C). The per-class structure of the gap and of the bias (Section III-D and Fig. 3) is consistent with a detector-quality bottleneck rather than a counter limitation.

Several limitations bound the scope of these findings. The benchmark uses a single dataset of 953 trees from two plantations in South Kalimantan, so absolute numbers may shift on other regions, varieties, or acquisition protocols. Only one detector checkpoint (YOLOv26-medium) is evaluated, and a different detector family could change the per-class error profile even if the headline gap remains. The tree-level counters treat each tree independently and ignore plantation-scale spatial or temporal regularities that operational forecasts sometimes exploit. Finally, the four-class B1 through B4 taxonomy admits visual ambiguity at the B2/B3 and B3/B4 boundaries, which places a soft cap on what any detector-plus-counter pipeline can ultimately achieve.

Future work should therefore prioritize improving detector recall on B3 and B4, calibrating detection confidence for counting use, and using explicit cross-view association or geometry-aware aggregation rather than expanding the tree-level feature bank further.

IV. Conclusion

This paper benchmarks multi-view tree-level oil palm FFB counting under a fixed detector by evaluating the same counters in a GT detection setting and a fixed-detector setting. Under GT detection, ElasticNet reaches 98.05% Class ±1 Acc and 92.20% Tree ±1 Acc on the 141-tree test split, showing that the tree-level counter is near ceiling when detections are accurate. Under the fixed YOLOv26-medium detector, the best Ridge + F_{all} pipeline reaches 77.48% Class ±1 Acc and 32.62% Tree ±1 Acc, and an eight-bank feature ablation closes only $\approx$1.4 percentage points of the gap. The remaining 20.57 percentage-point Class ±1 Acc gap is therefore a detector-quality bottleneck, not a counter limitation. The path to better BBC-style B1 to B4 counting on this dataset runs primarily through better detection, especially on B3 and B4, and through multi-view evidence handling that is explicitly designed around detector uncertainty.

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Code and Data Availability

Code: https://github.com/ULM-SawitMVC/Baseline-SawitMVC

Data: https://huggingface.co/datasets/ULM-DS-Lab/SawitMVC-YOLO

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